

# Renewable Energy Generation Forecasting Model for Reducing Power System Load and Solving Output Control Issues

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**Abstract**—Global population growth and technological advancements have led to a surge in energy consumption, resulting in substantial greenhouse gas emissions and severe climate change. In response, various countries have implemented climate policies to combat climate change, which have accelerated the development of renewable energy. While renewable energy is more environmentally friendly than conventional fossil fuel-based energy, it still faces numerous challenges due to the lack of technological advancement. This study aims to alleviate output control issues and reduce the burden on the power grid by forecasting renewable energy generation using meteorological data, generation data, and power demand data. Unlike conventional fossil fuel-based power facilities, which can adjust output in real-time, renewable energy facilities must forecast power generation to supply electricity effectively. Unchecked expansion of renewable energy facilities without considering power generation could lead to severe issues, including large-scale blackouts, beyond simple output control problems. This research employs the Long Short-Term Memory (LSTM) model, demonstrating through collected and analyzed data that solar and wind power generation can be effectively forecasted. By utilizing the developed renewable energy generation forecasting model and power demand data, this study can aid in forecasting curtailment and facilitate solutions such as optimal placement of Energy Storage Systems (ESS) and expansion of power grid infrastructure to mitigate grid load and address generation curtailment issues. Ultimately, it contributes not only to the effective achievement of Korea's CFI 2030, aimed at carbon neutrality, but also to the expansion of renewable energy facility deployment.

**Keywords**—Renewable Energy, Data Analysis, Output Control, Power Generation Forecasting, Energy Storage Systems

## I. INTRODUCTION

With the global population increase and advancements in digital innovation technologies, energy consumption has risen significantly [1]. Notably, the consumption of fossil fuel-based energy has increased further. The growing use of fossil fuels results in large-scale greenhouse gas emissions,

significantly impacting global warming and accelerating climate change [2]. According to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC), human activities, including greenhouse gas emissions, have raised the global surface temperature by 1.1 °C between 2011 and 2020, compared to the period from 1850 to 1900. Persistent greenhouse gas emissions are expected to escalate warming, leading to a 1.5 °C rise by 2040 in nearly all scenarios. Consequently, various countries worldwide are implementing diverse climate policies, which are increasingly linked to renewable energy [3-6].

Korea is also implementing various climate policies in response to the global trend of climate change. One such initiative is Jeju Island's Carbon Free Island (CFI) 2030 policy, which aims to achieve low-carbon green growth on Jeju Island while addressing the global issue of climate change. The goal of this policy is to supply 100% of the island's power demand with renewable energy by 2030, deploying a total of 4.085 GW of renewable energy, including 1.411 GW of solar power and 2.345 GW of wind power. However, unlike traditional fossil fuel-based energy, renewable energy generation is inconsistent, which can negatively impact power quality, stability, system operation, and control, and in the worst case, may even necessitate the cessation of power generation. Solar and wind power systems are highly influenced by weather conditions, leading to either low generation or excessive energy output beyond the grid's capacity, depending on the weather.

As a result, grid instability issues are also increasing in Jeju Island. According to the 10th Basic Plan for Long-term Electricity Supply and Demand by the Ministry of Trade, Industry, and Energy, the number of output control instances due to excess renewable energy generation in Jeju surged from three instances in 2015 to 132 instances in 2022, indicating a significant annual increase in the output control rate. While the expansion of renewable energy facilities effectively reduces carbon emissions, alternatives are necessary for the stable operation of the power grid.

This study aims to analyze meteorological data, renewable energy generation data, and power demand data to forecast renewable energy generation, thereby mitigating output control phenomena and enabling stable system operation. By forecasting power generation, it is possible to resolve output control issues of renewable energy facilities through various methods such as installing ESS, expanding High Voltage Direct Current (HVDC) infrastructure, or adjusting fossil fuel-based energy production. This will ensure the stable operation of renewable energy facilities and the power grid, contributing to the expansion of renewable energy deployment and ultimately addressing climate change through carbon neutrality.

In this paper, Chapter 2 reviews existing literature on power generation forecasting for solar and wind power systems. Chapter 3 provides an overview of the overall system and data used for generation forecasting, while Chapter 4 presents the results obtained using the proposed model. Chapter 5 concludes with a summary of the findings, limitations of existing research, and directions for future research.

## II. RELATED WORK

Solar and wind power generation systems are key components of renewable energy, and extensive research has been conducted in the field of power generation forecasting for these systems [7, 8]. Weather is the most significant factor affecting the performance of solar and wind power systems. For solar power systems, solar irradiance data has the greatest impact; however, other factors such as ambient temperature, wind speed, and relative humidity also significantly influence power generation by affecting the temperature of photovoltaic cells [9]. These meteorological data are highly variable over time, which is why the LSTM model was applied in this study.

In Reference [9], wind speed and ambient air temperature data, in addition to solar radiation data, were used in solar power generation forecasting models; however, the study employed ANFIS, RSM, and ANN models instead of the LSTM model. For wind power generation systems, wind speed is the most influential factor [10]. In Reference [11], a deep learning-based hybrid structure forecasting model was used to forecast wind power generation to improve forecasting accuracy, but it did not consider meteorological factors such as temperature, pressure, and humidity in the wind power generation forecasting experiments. This study aims to develop a solar and wind power generation forecasting model using meteorological data, existing power generation data, and the LSTM model.

Additionally, by forecasting curtailment through the developed model and using power demand data, this study seeks to reduce grid load and address curtailment issues in solar and wind power facilities through measures such as optimal ESS placement and power grid infrastructure expansion.

## III. METHODS

In Chapter 3, we describe the methodology used to forecast the power generation of solar and wind energy systems.

### A. Solar and Wind Power Generation Forecasting Model Architecture

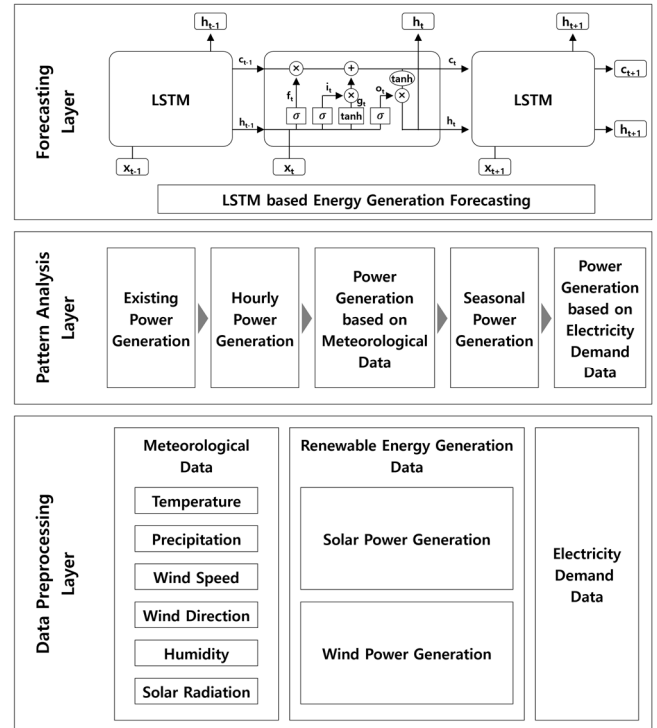


Fig. 1. 3-Layer Structured Solar and Wind Power Generation Forecasting Model Architecture

Fig. 1 illustrates the proposed 3-layer structured model for forecasting solar and wind power generation. In the first stage, meteorological data, renewable energy generation data, and power demand data required for forecasting are collected. In the second stage, the collected data is analyzed and processed to create data suitable for the LSTM model application. In the final stage, the LSTM model is applied to the data to enable the proposed model to forecast power generation.

### B. Data Collection and Preprocessing

The datasets used in this study are categorized into three types: energy generation data, meteorological data, and power demand data.

1) *Energy Generation Data*: Energy generation data is sourced from the Korea Power Exchange, specifically focusing on hourly solar and wind power generation data in the Jeju region. This data is based on the amount of energy generated through solar and wind power that was traded each hour. The dataset covers the period from January 1, 2023, to December 31, 2023, with records collected at hourly intervals.

2) *Meteorological Data*: Meteorological data is obtained from the Automated Synoptic Observing System (ASOS) provided by the Korea Meteorological Administration through its Open MET Data Portal. ASOS refers to ground observations conducted simultaneously at all observation sites at designated times to understand synoptic-scale weather. The data provided includes elements such as temperature, precipitation, wind speed, wind direction, humidity, solar radiation, sunshine duration, snow, cloud cover, visibility, surface condition, surface and subsurface temperature, weather phenomena, evaporation, and phenomenon codes. In

this study, we used only temperature, precipitation, wind speed, wind direction, humidity, and solar radiation data, which are deemed to influence solar and wind power generation. Additionally, meteorological data collected from areas within Jeju Island where solar and wind power facilities are concentrated was used. The data period is set to match the energy generation data, and hourly interval data was used.

3) *Power Demand Data*: Power demand data is sourced from the Korea Power Exchange, comprising hourly power demand data for the Jeju region. This dataset reflects the hourly demand performance of the generation units in Jeju, with the period also set to match that of the generation data.

For data preprocessing, outliers and missing values were handled for each dataset. After handling missing values, the data was arranged sequentially by time, merged, and then encoded.

### C. Data Pattern Analysis and LSTM Model Application

The LSTM (Long Short-Term Memory) model has advantageous features for handling and forecasting time-series data compared to other models. The reasons for using LSTM in this study are as follows:

#### 1) Consideration of Temporal Dependence

Both meteorological and generation data vary over time, and current values tend to depend on previous values. For example, wind power generation fluctuates with changes in wind speed, and solar power generation is greatly influenced by temporal changes in solar radiation. The LSTM model is designed to remember and learn from past data, allowing past values to influence current forecasting, which provides more accurate forecasting compared to other models.

#### 2) Learning Long-Term Dependencies

The LSTM model can maintain both long-term and short-term memory, allowing it to effectively learn from data over various time scales. This enables the model to forecast long-term trends in complex time-series data, such as weather patterns.

#### 3) Learning Repetitive Patterns

Power demand data exhibits repetitive patterns on daily, weekly, monthly, and seasonal scales. For instance, power consumption increases during summer and winter due to heating and cooling demand, and power usage is higher during the day compared to nighttime. The LSTM model can effectively learn from data with such repetitive patterns, making it well-suited for forecasting.

## IV. RESULTS

Fig. 2 and 3 illustrate the differences between the forecasted and actual power generation using the model. Fig. 2 shows the comparison between forecasted and actual solar power generation.

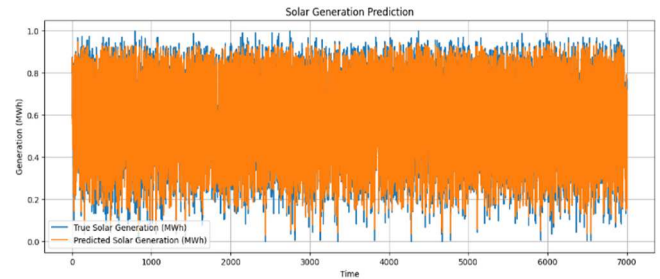


Fig. 2. Comparison of Forecasted and Actual Solar Power Generation in Jeju Island

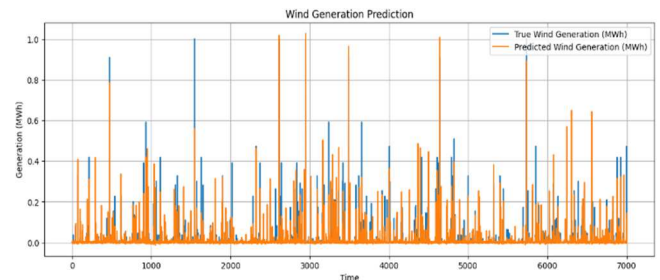


Fig. 3. Comparison of Forecasted and Actual Wind Power Generation in Jeju Island

As shown in TABLE I, the solar power generation forecasting exhibit high accuracy, with an MAE of approximately 0.02, RMSE of 0.03, and an  $R^2$  score of 0.98, indicating strong performance across all metrics.

TABLE I. ACCURACY METRICS FOR THE FORECASTING MODEL

| Evaluation Metrics | Solar    | Wind     |
|--------------------|----------|----------|
| MAE                | 0.024140 | 0.006953 |
| RMSE               | 0.033203 | 0.029026 |
| $R^2$ Score        | 0.978796 | 0.669726 |

Fig. 3 displays the comparison between forecasted and actual wind power generation. For wind power generation forecasting, the MAE and RMSE are approximately 0.01 and 0.03, respectively, but the  $R^2$  score is 0.67, indicating lower forecasting accuracy. This suggests that while the data used is effective for the solar power forecasting model, additional data acquisition or modifications to the model structure are necessary for improving the accuracy of the wind power forecasting model.

## V. CONCLUSION

In this study, we developed a model to forecast solar and wind power generation using existing solar and wind generation data, meteorological data, power demand data, and the LSTM model. The results show that the data and LSTM model perform well in forecasting solar power generation, but additional data or a different model is required for wind power generation.

Although this paper focused on power generation forecasting based on meteorological and power demand data, future research will aim to develop an output control forecasting algorithm by linking the data and model used in this study with output control frequency data recorded during

the same period. Additionally, further research will be conducted to improve the accuracy of the wind power forecasting model and to develop a forecasting algorithm for the charging and discharging rates of ESS connected with renewable energy generation.

By forecasting output control through power generation forecasting, this research can help prevent output control issues in renewable energy facilities, contributing significantly not only to the effective achievement of Korea's CFI 2030 plan but also to the future expansion of renewable energy deployment.

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